

YASH KUMAR

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B.Tech AI/ML student (2nd year, CGPA: 8.44/10) with hands-on experience building and deploying production-grade ML and LLM systems — from YOLOv8 computer vision pipelines (94.6% mAP) to agentic RAG systems with hybrid retrieval, LangGraph routing, and LLM-powered agents using RAG, LangChain, and Google ADK on GCP. Delivered 3 end-to-end projects spanning model training, FastAPI backends, and cloud deployment.

EDUCATION

B.Tech in Artificial Intelligence & Machine Learning

2024 – 2028 (Expected)

University School of Automation & Robotics, GGSIPU EDC | CGPA: 8.5 / 10 (4 semesters)

Delhi, India

Relevant Coursework: Machine Learning · Deep Learning · Computer Vision · Data Science · DSA · DBMS · Operating Systems · OOPs

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), TypeScript, SQL

ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, YOLOv8, OpenCV

AI & LLMs: LangChain, LangGraph, RAG (hybrid vector + BM25 retrieval), ChromaDB, Google ADK, Gemini, LLaMA-3, Groq

Backend & DB: FastAPI, Flask, Node.js, Express.js, PostgreSQL, MongoDB, Firebase

Frontend: React.js, Next.js, Tailwind CSS, Streamlit

Cloud & DevOps: Google Cloud Platform (GCP), Cloud Run, Docker, Git, Vercel, Render, Linux

KEY PROJECTS

QueDB — AI-Powered NL-to-SQL Engine | quedb.vercel.app

2025

FastAPI · React · LangChain · LangGraph · LLaMA-3 70B (Groq) · ChromaDB · PostgreSQL · JWT Auth

- Engineered full-stack AI query engine converting natural language to executable SQL across PostgreSQL, MySQL, and SQLite using LLaMA-3 70B — enabling non-technical users to query databases without writing SQL.
- Built a RAG pipeline (ChromaDB + sentence-transformers) for schema-aware context injection, plus a multi-retry self-correction loop that feeds SQL errors back to the LLM to auto-recover from failed queries.
- Designed live Schema Explorer with Monaco code editor for interactive query editing — deployed on Railway (backend) + Vercel (frontend).

RepoSage — Production Codebase RAG Agent | reposage-lake.vercel.app

2026

FastAPI · React · TypeScript · LangGraph · ChromaDB · BM25 · tree-sitter · Groq (LLaMA-3 / GPT-OSS-120B)

- Built a production-deployed RAG agent that answers natural-language questions about any indexed GitHub repository, with every claim in the answer cited back to an exact file and line range.
- Parsed repositories with tree-sitter into function- and class-level chunks, then indexed them with hybrid retrieval — ChromaDB vector search fused with BM25 via Reciprocal Rank Fusion, plus an optional cross-encoder reranking stage.
- Designed a LangGraph agentic router (direct lookup / multi-hop decomposition / clarification) and a live indexing pipeline (POST /index) so users can point the tool at any public repo at runtime — deployed with FastAPI (Render) + React/TypeScript (Vercel).

Helmet Detection & Traffic Monitoring System | helmet-detection-project.vercel.app

2025

Python · YOLOv8 · OpenCV · FastAPI · React · TypeScript · Render · Vercel

- Built real-time helmet compliance detection system using YOLOv8 (94.6% mAP@0.5, 73.9% mAP@0.5:0.95, ~5ms inference) with an overlap-based person-motorcycle matching pipeline and multi-stage detection via a FastAPI REST backend.
- Deployed end-to-end: model training → FastAPI (Render) → React/TypeScript frontend (Vercel) with auto-generated Swagger documentation.

EXPERIENCE

ML Intern — FlyRank

Jul 2026 – Aug 2026

Applied Search Intelligence — Google Search Ranking & Discoverability | Remote

Worked the Refresh / Content Opportunity Scoring lane: built a time-aware Random Forest classifier predicting content growth/decline/review status on FlyRank's production search-ranking warehouse, improving F1-macro from a 0.208 baseline to 0.380.

Open Source Contributor — GirlScript Summer of Code 2026

May 2026 – Present

AI Agents Track | Remote

Contributing to open-source AI/ML repositories under the competitive AI Agents Track, collaborating on Python and TypeScript-based agentic systems through issue resolution and pull requests.

CERTIFICATIONS

Google Cloud Gen AI Academy APAC 2026 — Google Cloud & Hack2skill(Cohort 1 & 2)

Apr – Jul 2026

IBM AI Fundamentals — IBM SkillsBuild

May 2026

Deloitte Data Analytics Job Simulation — Forage

Mar 2026

ACHIEVEMENTS & ACTIVITIES

- Smart India Hackathon 2025 (SIH) — Certificate of Appreciation; cleared University-Level 2 rounds of India's largest national hackathon.
- Google Cloud Gen AI Academy APAC 2026 — Selected for Cohort 1 and Cohort 2; built and deployed live AI agents on Google Cloud Run as part of the competition.
- CodeX Hackathon — Developed and presented an AI project under competitive hackathon conditions.